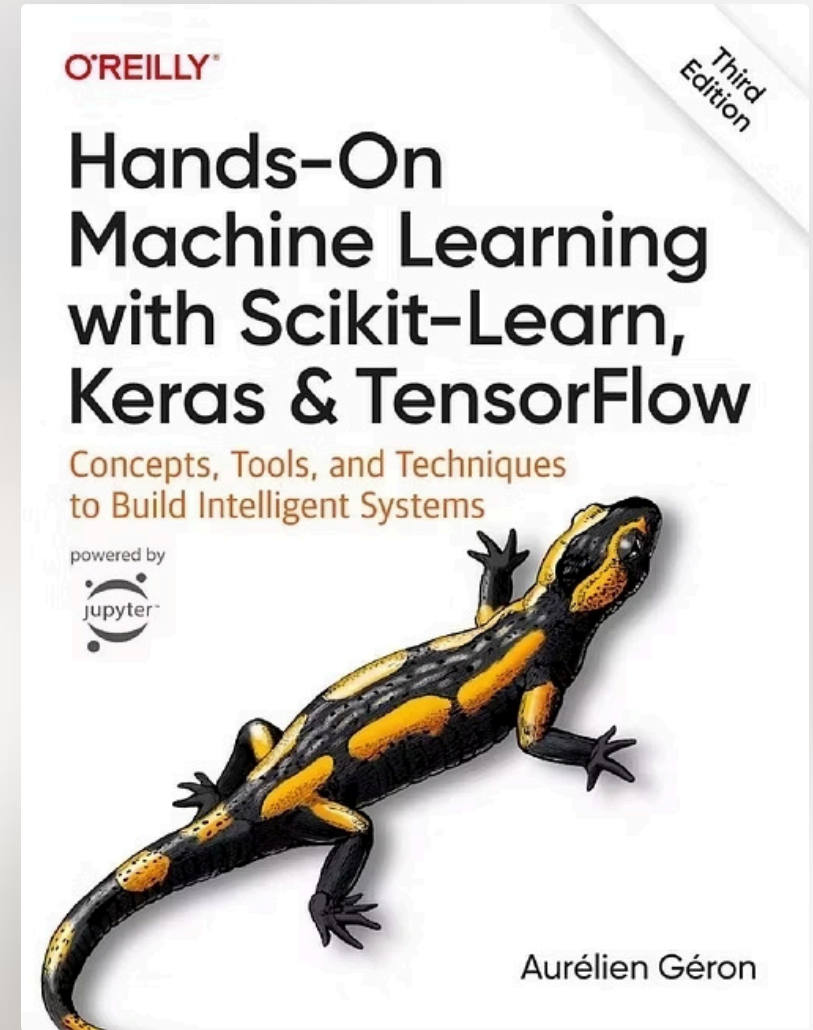


Machine Learning

Lessons based on the Textbook "Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow" by Aurélien Géron, 2023.

RP by Rodrigo Martins Pagliares

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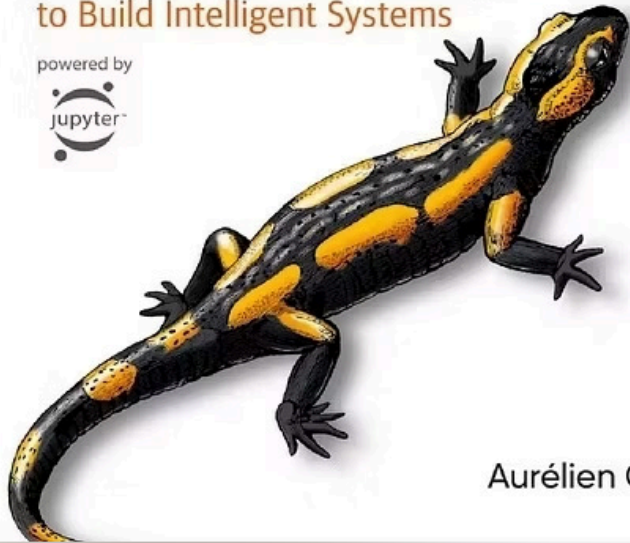
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Third
Edition

Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow

Concepts, Tools, and Techniques
to Build Intelligent Systems

powered by



Aurélien Géron

Machine Learning

Chapter 1 – The Machine Learning Landscape



Agenda for Chapter 1

1. What Is Machine Learning?
2. Why Use Machine Learning?
3. Examples of Applications
4. **Types of Machine Learning Systems**
5. Main Challenges of Machine Learning
6. Testing and Validating

Agenda for Chapter 1

1. What Is Machine Learning?
2. Why Use Machine Learning?
3. Examples of Applications
4. **Types of Machine Learning Systems**
 - Classification criteria #1: Supervision during training
 - Supervised, Unsupervised, Semi-Supervised, Self-Supervised, and Reinforcement Learning
 - Classification criteria #2: Approach to Learning Method
 - Batch and Online Learning
 - Classification criteria #3: Learning Method
 - Instance-Based and Model-Based Learning
5. Main Challenges of Machine Learning
 - Insufficient Quantity of Training Data
 - Non-representative Training Data
 - Poor-Quality Data
 - Irrelevant Features
 - Overfitting the Training Data
 - Underfitting the Training Data
 - Stepping Back
6. Testing and Validating
 - Hyperparameter Tuning and Model Selection
 - Data Mismatch



Classification criteria #2:

Approach to Learning

Method

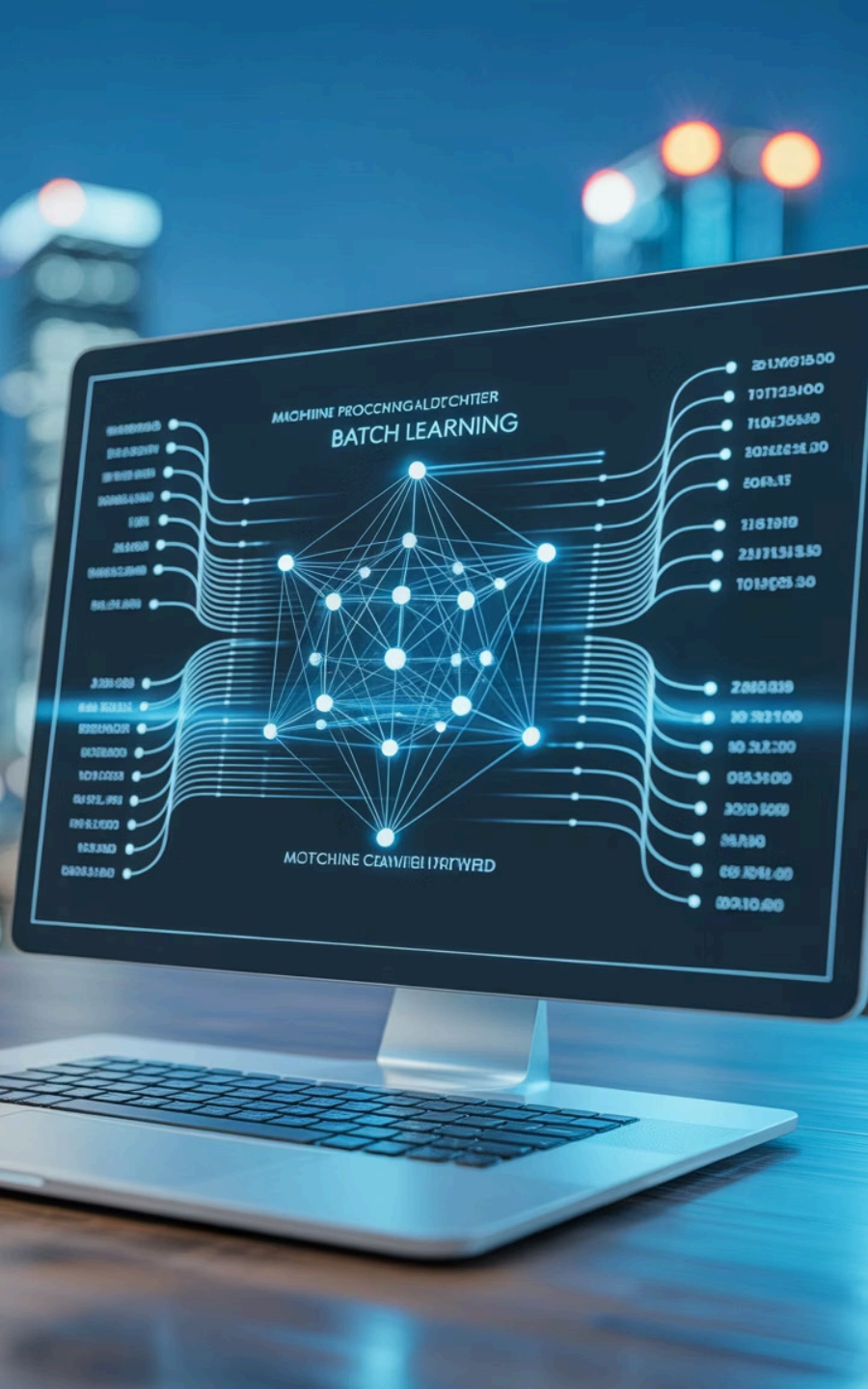
Batch and Online Learning

Batch Learning: Overview

- The system is trained using all available data at once.
- Learning is not incremental.
 - The model DOES NOT update after deployment.

Training Process:

- The system is typically trained offline since training will generally take a lot of time and computing resources.
- After training the system is deployed in production to apply the learned model.



Challenges of Batch Learning

Slow and Resource-Intensive:

- Training on large datasets can take significant time and computing resources.
- The model must be retrained periodically as the world evolves.

Model Decay:

- The model's performance can degrade over time due to data drift or model rot.
- Data drift/model rot: Changes in the data distribution over time.



When to Retrain a Batch Learning Model



Slow Decay

For relatively **stable data**, like images of cats and dogs, **retraining** can be done **less frequently**.



Fast Decay

For **rapidly evolving data**, like financial market predictions, more **frequent retraining** is **necessary**.

The **frequency of retraining** depends on the **specific use case** and how **quickly** the underlying **data patterns** change.





Warning

- Even a model trained to classify pictures of cats and dogs may need to be retrained regularly
 - Not because **cats and dogs will mutate overnight**, but because **cameras keep changing, along with image formats, sharpness, brightness, and size ratios**

Retraining Batch Models



Retraining Process

- **Retrain the model on the full dataset** (old + new data) to update it.
- **Offline learning:** The model **DOES NOT** learn once deployed; It just **applies** what was learned.

Automation

The process of **training, evaluating, and launching** can be **automated** for efficiency.

Limitations of Batch Learning

High Resource Consumption:

- **Retraining the entire dataset** can require extensive computing resources (CPU, memory, storage)
- **Costly:** Training with large datasets regularly can become expensive

Slow Adaptation to Change:

- **Reactive approach:** It may take **hours or days to retrain**, which is not ideal for fast-evolving data



Alternatives to Batch Learning

Incremental Learning:

- For **systems with limited resources** or **fast-changing data**, incremental learning algorithms are preferred.
- **Example:** Models that can **learn from new data on the fly**, such as in **mobile devices** or **Mars rovers**.



Mobile Devices

Limited resources require efficient learning approaches



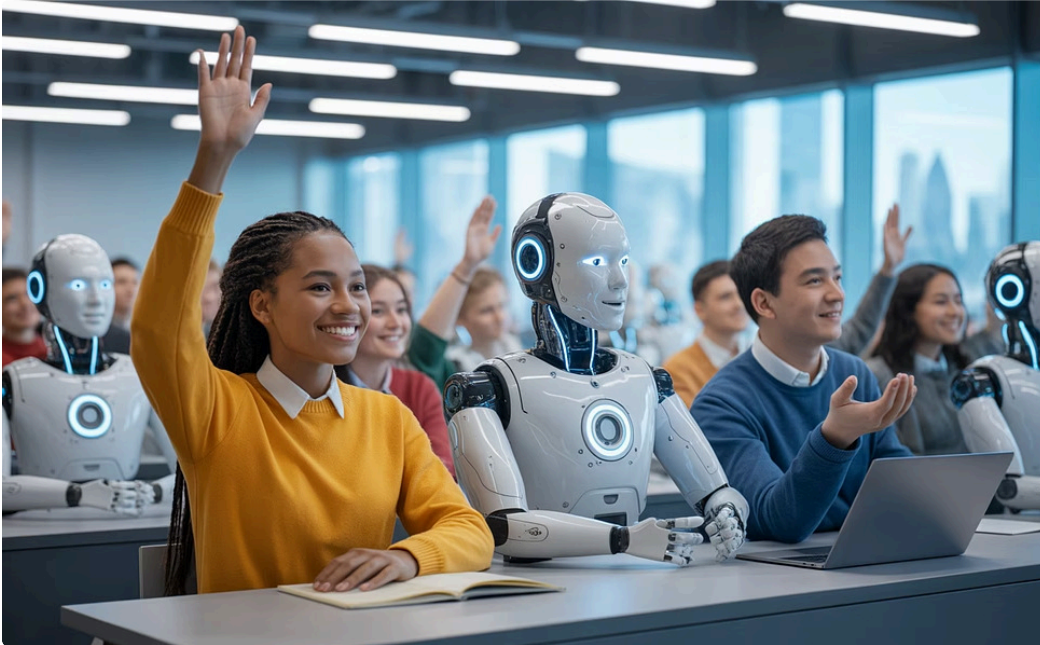
Mars Rovers

Remote operation requires **autonomous learning** capabilities

Next: Online Learning

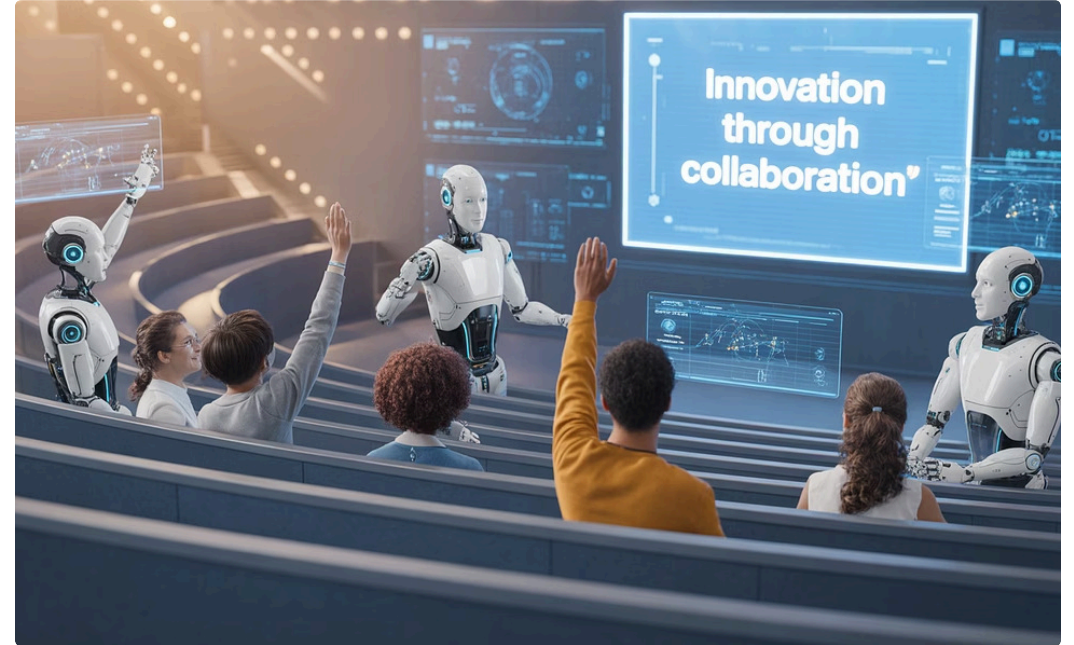


Q&A



Open Floor

Open floor for discussion or clarifications



Discussion

Share your thoughts and questions about AI concepts